Semantic Retrieval of Aviation Safety Reports with Frozen Sentence Embeddings

Jacob Tang Aarambh Shrestha
CIS 634, Natural Language Processing Dr. Mansi Bhavsar 21 September 2026

1 Research paper summary

1.1 Bibliographical information

Nils Reimers and Iryna Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," *Proceedings of EMNLP-IJCNLP*, pp. 3982–3992, Hong Kong, 2019. `https://doi.org/10.18653/v1/D19-1410`

1.2 Background

Before this paper, the most accurate way to judge whether two sentences mean the same thing was to feed both into BERT together. That must be repeated for every pair: finding the most similar pair among 10,000 sentences takes about 50 million BERT runs, roughly 65 hours on a GPU, far too slow to search an archive. The obvious shortcut, using BERT's output for one sentence as its "meaning vector," does worse than simply averaging GloVe word vectors. The authors wanted sentence vectors that are fast to compare and still meaningful.

1.3 Contributions

Sentence-BERT (SBERT) fine-tunes BERT so that every sentence becomes one fixed-length vector, and sentences with similar meaning end up close together (by cosine similarity). It is trained on about one million sentence pairs labeled as agreeing, contradicting, or unrelated (the SNLI and MultiNLI datasets), using two copies of the same network that share weights (a "Siamese" setup). After training, each sentence is encoded once and a search becomes a quick vector comparison, so the 65-hour search drops to about 5 seconds. On seven standard sentence-similarity benchmarks, SBERT averages 74.9 (Spearman correlation $\times 100$), against 54.8 for averaged BERT outputs, 61.3 for GloVe, and 71.2 for Google's Universal Sentence Encoder. Its code became the `sentence-transformers` library we use.

1.4 Limitations

The speed result is convincing; the accuracy evidence is weaker outside general English. On an argument-similarity task with a topic unseen in training, SBERT falls to 50.7 while the slower pairwise BERT scores 57.2, and that is the setting closest to ours: a model applied to text unlike its training data. All test data is everyday English (news, image captions, forums), with no jargon, abbreviations, or numbers. Crucially for us, the paper never checks whether the vectors separate sentences that share words but differ in meaning, such as "did not descend" versus "descended." Squeezing a passage into one vector can blur exactly these details. The paper is convincing as a fast, practical tool, but leaves open whether it can be trusted on specialized text where small details matter, which is what our project tests.

1.5 Why this paper

SBERT is the method our project applies directly: we encode each safety report once and search by cosine similarity. We were drawn to whether a general-purpose language tool holds up in a safety-critical field where one word can reverse a report's meaning.

1.6 Wider research context

The paper connects to two core course themes: how to represent the meaning of a whole sentence as a vector, and why language is hard for computers (negation, word order, and numbers change meaning while barely changing the words). SBERT's "encode once, compare fast" design became the basis of modern dense retrieval; later models such as E5 train the same kind of encoder on far more data [5]. In aviation, Gao et al. built a semantic encoder to classify and retrieve safety reports [6], and Aviation-BERT and SafeAeroBERT pretrain BERT on aviation text [7, 8]. The field is active, but we found no public test of whether off-the-shelf embeddings keep details like negation and direction in pilot reports.

2 Project description

2.1 Goal and research question

Can a frozen sentence embedding retrieve NASA ASRS reports that describe the same operational issue when the wording differs, and does that ranking fail under negation, altitude, direction, or a change in event order?

In plain terms: when a safety analyst describes a problem in their own words, can a ready-made model find past reports about the same problem even if the pilots worded it differently, and does it get fooled by reports that use the same words but describe a different event? "Frozen" means we use the published models as is, with no extra training: we apply SBERT-style embeddings to a specialized domain.

2.2 Why this topic is worth studying

- **A real safety resource that is hard to search.** NASA's Aviation Safety Reporting System (ASRS) collects voluntary reports from pilots, controllers, and mechanics so the industry can spot hazards before they cause accidents. The ten-year public slice we use alone holds 47,723 reports, about 13 million words, far more than anyone can read.
- **Keyword search misses paraphrases.** One pilot writes "we climbed through our assigned altitude," another "overshot the level-off." Keyword search treats these as unrelated, while a common word like "altitude" returns thousands of off-topic hits. Embeddings are designed to match meaning, not exact words.
- **It tests embeddings where they are weakest and aviation is strictest.** Embeddings are known to struggle with negation, numbers, and word order, and in aviation those details are the whole story. A tool that ranks "did not descend" next to "descended" hands an analyst confidently wrong precedents, so we measure this risk directly instead of reporting only an average score.
- **Cheap, feasible, and useful either way.** The data is public and already downloaded, and frozen models need no training and can index every report in minutes on one GPU or Google Colab. If they work, any safety team could adopt them; if they fail, we show exactly

where, and Extension D tests whether a small fine-tune fixes it.

2.3 Task

Input: a short description of an operational issue, e.g., "The crew followed an incorrect altitude because two clearances sounded similar." **Output:** a ranked list of ASRS reports, with the passage that matched best. Each result is graded *2* (same issue, e.g., an altitude read-back mistake), *1* (related, e.g., an altitude error with a different cause), or *0* (shares only an airport, aircraft, or the word "altitude").

2.4 Data

We use the Hugging Face dataset `elihoole/asrs-aviation-reports` [2] (Apache-2.0): 47,723 de-identified ASRS reports from January 2012 to March 2022, pre-split into 38,655 training, 4,295 validation, and 4,773 test records. Each report has the reporter's free-text narrative (about 280 words on average) plus NASA analyst fields such as event type, flight phase, and a synopsis. We search the narrative only. The analyst fields help us pick a labeled sample of 2,000–3,000 reports focused on altitude, clearance, and communication problems, while all 47,723 reports stay searchable so wrong answers are realistic. Preprocessing is light: split long narratives into paragraphs (the models read a limited length), tag each piece with its report ID, and keep reports of the same incident in the same split.

2.5 Neural methods: what we reuse and what we build

Reused: two pretrained embedding models from Hugging Face, run with `sentence-transformers`: the small SBERT-family model `all-MiniLM-L6-v2` [4] and the newer retrieval model `e5-small-v2` [5]. **Built by us:** paragraph splitting and report scoring (best paragraph versus whole report), the search index, 50–80 test queries with graded relevance labels, the evaluation code, and a *meaning-change test set* that is the original part of the project. Each test item has an original sentence, a paraphrase with the same meaning, and a version where one small edit changes the meaning; the model passes if it scores the paraphrase as closer.

Change type	Original	Same meaning	Changed meaning
Negation	The crew did not descend	The crew maintained altitude	The crew descended
Altitude	Descend to six thousand feet	Reduce altitude to 6,000 ft	Descend to four thousand feet
Direction	Turn left heading 240	Make a left turn to heading 240	Turn right heading 240
Event order	The aircraft leveled off, then turned	After leveling off, it turned	The aircraft turned, then leveled off

2.6 Baselines and evaluation

We compare four systems on the same data: **BM25** keyword search (a standard library implementation, standing in for today's keyword tools), frozen **MiniLM**, frozen **E5**, and a **hybrid** that merges the BM25 and embedding rankings. Our main metrics are **Precision@10** (of the top ten results an analyst sees, how many are about the same issue) and **nDCG@10** (which also rewards putting the best matches first and gives partial credit to grade-1 results). For the meaning-change test we report the percentage of items passed, separately for negation, altitude, direction, and event order; 50% is chance. No published scores exist for this exact task, so we compare our four systems (for reference, SBERT averages 74.9 on general sentence similarity [1]). Both teammates label results blind to the system, double-labeling part of the set to check agreement. We expect embeddings to beat keyword search on paraphrases but fail more often on negation and event order. ASRS reports are voluntary and unverified, so results say nothing about how often incidents happen [3].

2.7 Stretch goals (only after the core comparison is done)

- **B. Which version of a report searches best?** Repeat the evaluation on three versions of each report: the raw narrative, a short standardized event summary written with a fixed vocabulary ("read-back matched the wrong altitude; the aircraft then descended"), and the analyst's structured fields written out as text. This shows whether NASA's existing metadata does the job as well as the narrative.
- **C. Searching flight data with language.** Apply the same queries to short segments of public ADS-B flight tracks from OpenSky, such as "a descending left turn followed by level flight," linking written reports to recorded flight paths. This depends on obtaining a small sample, and tracks cannot show *why* a pilot did something.
- **D. Small fine-tune.** If the frozen models miss important distinctions, fine-tune MiniLM or E5-small on a few hundred labeled report pairs, including tricky near-miss pairs, on one 12 GB GPU, and test only on queries held out from training. This tells us whether a small, cheap adaptation is worth it or the off-the-shelf model is good enough.

2.8 Timeline

26 Oct milestone: data prepared, BM25 and both frozen models scored, first labeled queries and meaning-change items. **23 Nov talk:** four-system comparison and failure examples. **30 Nov report:** full results, plus Extension B, then D, then C as time allows.

References

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